# KANTIPUR ENGINEERING COLLEGE

*(Affiliated to Tribhuvan University, IOE)*

**DHAPAKHEL, LALITPUR**

**[SUBJECT CODE: CT654]**

**A MINOR PROJECT FINAL REPORT ON**

# MUHAAR : A FACIAL SKIN TYPE DETECTION TOOL


**SUBMITTED BY:**

**Payal Singh Yadav [37044]**

**Richa Bhatta [37055]**

**Safal Khatiwada [37058]**

**Swati Mahato [37078]**


**SUBMITTED TO:**

**DEPARTMENT OF COMPUTER AND ELECTRONICS ENGINEERING**

**IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN COMPUTER ENGINEERING**

**MAR, 2026**

**MUHAAR : A FACIAL SKIN TYPE DETECTION TOOL**


**SUBMITTED BY:**

**Payal Singh Yadav [37044]**

**Richa Bhatta [37055]**

**Safal Khatiwada [37058]**

**Swati Mahato [37078]**

**SUBMITTED TO:**

**DEPARTMENT OF COMPUTER AND ELECTRONICS ENGINEERING**

**KANTIPUR ENGINEERING COLLEGE**

**DHAPAKHEL, LALITPUR**


**IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR IN COMPUTER ENGINEERING**

**MAR, 2026**

# ACKNOWLEDGMENT

We would like to express our gratitude to everyone who helped us complete this project. First and foremost, we would like to acknowledge the crucial role of our teachers in the Department of Electronics and Computer Engineering for their guidance, support, and feedback throughout the project. We extend our sincere appreciation to Er. Rabindra Khati, Head of the Department of Computer and Electronics, for his valuable support. Additionally, we are grateful to Er. Pralhad Chapagain , Er. Bishal Thapa , Er. Nirajan Acharya and Er. Aliz Shrestha for their continuous help and encouragement during the course of our project.

Next, we would like to express our gratitude to our classmates for providing constructive feedback and engaging in thought-provoking discussions regarding our project. We also extend our thanks to all the lecturers of our department and our institution, Kantipur Engineering College, for providing the resources and environment necessary to conduct and guide this research from the beginning to the completion of this project.

Finally, a special gratitude to our family and friends for their love, encouragement, and support throughout our academic journey.

Thank you all for your invaluable contributions to this project.

Payal Singh Yadav [37044]
Richa Bhatta [37055]
Safal Khatiwada [37058]
Swati Mahato [37078]

# ABSTRACT

Understanding skin type is fundamental to effective skincare and the prevention of dermatological issues. To support this, a system is needed to identify and analyze facial skin type. This project introduces "MUHAAR: A Facial Skin Type Detection Tool", an automated system to classify human facial skin into three categories: dry, normal, and oily.

Muhaar utilizes a Convolutional Neural Network (CNN) based on the MobileNetV2 architecture, leveraging transfer learning from the ImageNet dataset. To ensure high performance, a dataset sourced from Kaggle underwent rigorous preprocessing, including normalization and data augmentation. By integrating a custom classification head with Global Average Pooling and Dropout layers, the system effectively extracts complex skin texture patterns. Preliminary results demonstrate a validation accuracy of 93.09%, with balanced precision and recall across all classes. This tool offers a scalable, objective approach to skin analysis, with future work focusing on fine-tuning the model for enhanced real-world application.

This study highlights the use of AI in facial skin type detection and its benefits in dermatology and cosmetology.

**Keywords**- *CNN, Normalization, Global Average Pooling, Dropout, Precision, Recall*

# TABLE OF CONTENTS

# LIST OF FIGURES

# LIST OF TABLES

# LIST OF ABBREVIATIONS

**Adam** – Adaptive Moment Estimation (optimizer)

**AI –** Artificial Intelligence

**BSTI –** Baumann Skin Type Indicator

**CNN –** Convolutional Neural Network

**DO –** Dry/Oily

**FP –** False Positive

**FN –**False Negative

**GAP –** Global Average Pooling

**Google Collab –** Google Collaboratory

**GPU** – Graphics Processing Unit

**IDE –** Integrated Development Environment

**LSTM –** Long Short-Term Memory

**NumPy** – Numerical Python

**OpenCV** – Open-Source Computer Vision Library

**OS –** Operating System

**PN –** Pigmented/Non-pigmented

**RAM –** Random Access Memory

**ReLU** – Rectified Linear Unit (activation function)

**RMSProp –** Root Mean Square Propagation (optimizer)

**ROI –** Region of Interest

**SGD** – Stochastic Gradient Descent (optimizer)

**SR** – Sensitive/Resistant

**SSD –** Solid State Drive

**TP –**True Positive

**TN –**True Negative

**UV –** Ultraviolet

**WT –** Wrinkle-prone/Tight

# CHAPTER 1
# INTRODUCTION

## 1.1. Background

Skin is the biggest organ of the body and one of the most crucial organs to guard us against the threats of the environment, such as pollution, toxic UV light, and bacteria. It also helps in regulating body temperature, preventing dehydration, and serving as a protective cover between our inner system and outer world. Since it's so exposed, our skin is susceptible to several conditions such as acne, eczema, premature aging, or dryness, most of which are dependent on our skin type.

Choosing the perfect skincare products depends largely on knowing your skin type. For example, individuals with dry skin would require products containing moisturizing agents such as hyaluronic acid, glycerin, and ceramides to restore lost moisture and strengthen the skin barrier. Individuals with oily skin, on the contrary, require light, oil-free products that can control excess oil and prevent pore blockages such as salicylic acid or niacinamide.

This is where skin type detection comes into play. Through exactly identifying whether one has dry, oily, or combination skin, skin care products and routines can then be personalized to meet the skin's individual needs. Computerized detection systems, powered by computer vision and AI, allow this to be processed at a quicker pace, with convenience, and in accessibility, especially for those who may not be able to take advantage of dermatological consultation. Ultimately, skin type recognition allows individuals to make educated decisions about their skincare, leading to better results and improved confidence in their daily practice.

## 1.2. Problem Statement

The most exposed and sensitive part of skin is the face, most people continue to ignore its well-being and health. The daily routine subjects’ skin to pollution, harsh climate, stress, and dehydration, but individuals barely pay attention to how these factors affect their facial skin. This absence of information is potentially causing widespread but possibly serious issues such as acne, dryness, premature aging, infection, and irregular skin tone. One of the root causes of this growing phenomenon is the lack of information about one's own skin type and how it reacts to changes in the environment. Since one does not know what specific needs their skin has, they are unable to take the right steps towards its health, only realizing the damage when problems are apparent or aggravating. Ignorance not only impacts how one looks, but it can also ruin self-confidence and overall well-being.

Therefore, a system that provides knowledge about basic skin type is necessary and crucial for preventing long-term skin issues and encouraging healthy skin.

## 1.3. Objective

i. To develop a model that classifies the human skin type in categories like dry, normal, oily using image-based analysis.
ii. To reduce dependency on visual guess work for skin type analysis which is often inaccurate.

## 1.4. Application and Scope

i. It can be used various skin care app for skin care recommendations
ii. It acts as a dermatology virtual assistant to pre assess skin type
iii. It can be used by professionals before facials, peels, or laser treatments for spa.
iv. It can be used by dermatology students who are learning in skincare-related fields by providing data.
v. It allows influencers to use image-based analysis to validate product use cases

## 1.5. Features

i. Accurately detect the skin type as dry, normal, oily
ii. Eliminates the need for physical tests instead users can simply upload a photo for skin type detection
iii. Designed with modularity to allow future feature upgrades like acne detection.
iv. It is designed in way that it can easily integrate into web or mobile apps related to skin care

## 1.6. Feasibility Study

Before implementing the project design, conducting a feasibility analysis was essential. This analysis provided insight on how the project will perform and its potential impact in real-world scenarios, making it a crucial step before proceeding further.

### 1.6.1. Economic Feasibility

This determines whether the doing of the project is economically beneficial or not. Development will be carried out using existing hardware and standard workstations, avoiding any additional equipment expenses. This will significantly reduce the overall budget required for implementation. The cost-effectiveness of the chosen tools ensures that the system remains accessible and affordable for future development. Moreover, updates and improvements will be integrated over time without imposing extra financial strain, making the solution sustainable in the long run.

### 1.6.2. Technical Feasibility

The project will be technically feasible because the necessary software and tools such as: Vs Code, JavaScript, TensorFlow, Keras etc. will be readily available and could be downloaded from various online resources at no cost. Also, this system can adopt technological upgrades as it is developed under the consideration of software engineering principles.

### 1.6.3. Operational Feasibility

The system is user-friendly and compatible with standard phone or laptop cameras, requiring no special equipment. It offers an easy setup where users can simply upload or capture a photo to receive immediate results. Designed for broad accessibility, the project serves general users, skincare enthusiasts, and professionals. The system is scalable, with the potential to expand its capabilities to detect various skin issues. It delivers accurate and reliable analysis through advanced image processing techniques

### 1.6.4. Schedule Feasibility

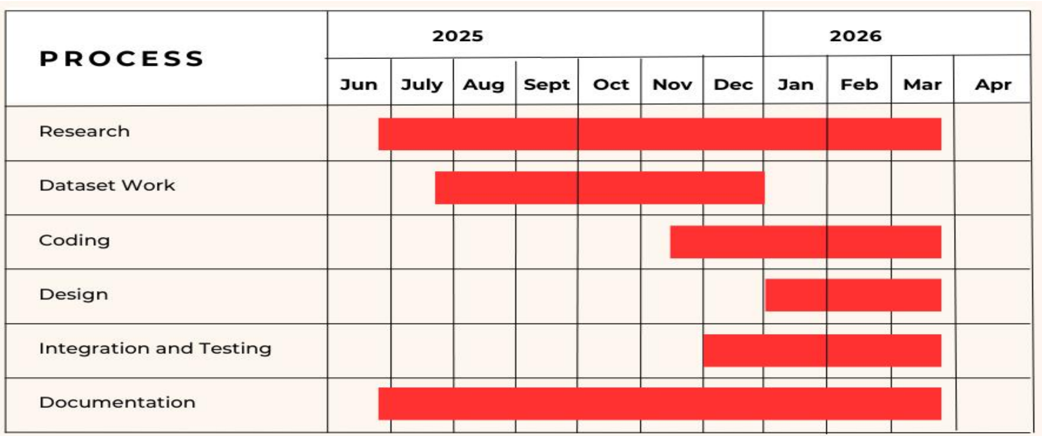


Fig 1.1: Gantt Chart

The project followed a team-based workload distribution, with tasks strategically divided among members. An incremental development approach was adopted, allowing the system to be built into module by module. Each member was responsible for specific components aligned with the project timeline. Regular progress tracking and check-ins ensured the project stayed on schedule, resulting in the timely completion of all major deliverables.

## 1.7. System Requirements

### 1.7.1. Development Requirements

Table 1.1: Hardware Requirements

| Components | Minimum Requirements |
|---|---|
| Processor | Intel i5 $7^{th}$ Generation |
| RAM | 8 GB |
| Storage | 256 GB SSD |
| GPU | Intel UHD/AMD Vega |

Table 1.2: Software Requirements

| Components | Minimum Requirements |
|---|---|
| Operating System | Windows 10, MacOS, Linux |
| Language | Python, JavaScript |
| Design Tools | Draw.io, Canva |

| | |
|---|---|
| ML libraries | TensorFlow, Keras, NumPy |

### 1.7.2. Deployment Requirements

Table 1.3: Hardware Requirements

| **Components** | **Minimum Requirements** |
|---|---|
| Processor | Dual Core (2 vCPU) |
| RAM | 2-4 GB |
| Storage | 10-20 GB SSD |

Table 1.4: Software Requirements

| **Components** | **Minimum Requirements** |
|---|---|
| Operating System | Windows 7, MacOS, Linux |
| Internet | 1-2Mbps |

# CHAPTER 2
# LITERATURE REVIEW

## 2.1. Related Search

The paper by Jianghong Ran et al. [1] proposes an automatic skin type measurement system based on an improved Inception-v3 deep learning model with transfer learning. The system classifies skin types using the four-dimensional BSTI framework such as dry/oily (DO), sensitive/resistant (SR), pigmented/non pigmented (PN), and wrinkle-prone/tight (WT). Training was performed on a multimodal dataset from 54 healthy Chinese volunteers, including physiological skin indicators, BSTI questionnaire responses, and facial images taken under three lighting conditions. Images were preprocessed using Gaussian filtering and region-of-interest extraction to enhance relevant features. The model achieved high accuracy: 91.11% (DO), 81.13% (SR), 91.72% (PN), and 74.9% (WT), surpassing traditional and previous deep learning methods. However, limitations include the small, demographically narrow dataset and potential variability from lighting conditions. Future improvements should focus on expanding the dataset of diverse populations. This method provides a more objective and efficient alternative for skin type assessment in dermatology and cosmetics.

The study by Sirawit Saiwaeo et al. [3] proposed a fine-tuned CNN architecture specifically designed for human skin type classification, categorizing skin into normal, oily, and dry types. Initially, well-known pre-trained models such as AlexNet, VGG16, and ResNet50 were evaluated using transfer learning but showed suboptimal performance for this task. The researchers compiled a dataset of 3,152 high-quality frontal facial images from individuals aged 20 to 60, with balanced representation across the three skin types. This dataset was split into training, validation, and testing subsets and augmented with techniques including rotation, scaling, flipping, and color adjustment to improve model robustness. During evaluation, using a new unseen dataset reported an accuracy of 89.70% (21.68% loss. Despite these promising results, the model has limitations such as potential overfitting due to the relatively modest dataset size and limited diversity, as well as challenges in generalizing to skin types or conditions not represented in the training data.

The paper by Ruchika Chouhan et al. [2] presents a deep learning approach for classifying human skin types to improve cosmetic product recommendations. The study utilizes a substantial dataset of 3,152 frontal facial images of individuals aged 20 to 60, categorized into normal (1,274 images), oily (1,120 images), and dry (758 images) skin types, and further augmented through techniques such as rotation, scaling, flipping, and color adjustment to expand the dataset to 3,408 images. Several well-known transfers learning models, including AlexNet, VGG16, and ResNet50, were evaluated, but the best results were achieved by a custom fine-tuned CNN architecture model features a structured image processing pipeline: images are imported from various devices, preprocessed and enhanced, and then classified using the CNN, which extracts the features for accurate skin type identification. The system not only classifies skin types but is also designed to recommend suitable cosmetic products based on the classification. The fine-tuned CNN achieved a validation accuracy of 99.62% with an average loss of 22.74%, and after further hyperparameter tuning, the accuracy reached 94.57% with a validation loss of 0.0113, demonstrating the model's high effectiveness and potential for real-world cosmetic and dermatological applications.

Fatma et al. [4] with other researchers who examine deep learning methods for skin type classification using clinical information. Compare CNN and LSTM models with hyperparameters being optimized as activation functions (Tanh, ReLU, Sigmoid), optimizers (Adam, SGD, RMSProp), and training epochs (50, 100, 500). Data was split between training (7,887), validation (1,126), and testing (2,253). CNN using Adam optimizer, Sigmoid activation, and 500 epochs performed the best with 83.75% accuracy, beating LSTM and other configurations, showing CNN's capability for this nonlinear classification task. Limitations, however, are that there is a single data source, lack of model interpretability analysis, and no comparison with dermatologists.

This research presents an AI-powered system for personalized skincare recommendations by combining ingredient analysis and facial skin assessment. Jinhee Lee et al. [5] uses a Transformer encoder to analyze cosmetic ingredient sequences and predict effectiveness for six skin concerns, while a modified U-Net processes facial images to detect pores, redness, acne, wrinkles, pigmentation, and dark circles. The system was trained on 8,000 curated cosmetic products with ingredient lists and mapped marketing tags, and annotated facial images from the LUMINI Kiosk V2. Results showed strong performance in matching products to skin concerns. However, limitations include unknown ingredient quantities, a relatively small dataset. Overall, the study shows promise for AI-driven skincare, but emphasizes the need for richer, more diverse data to improve accuracy and reliability. [5]

# CHAPTER 3
# METHODOLOGY

## 3.1 Model Training Process

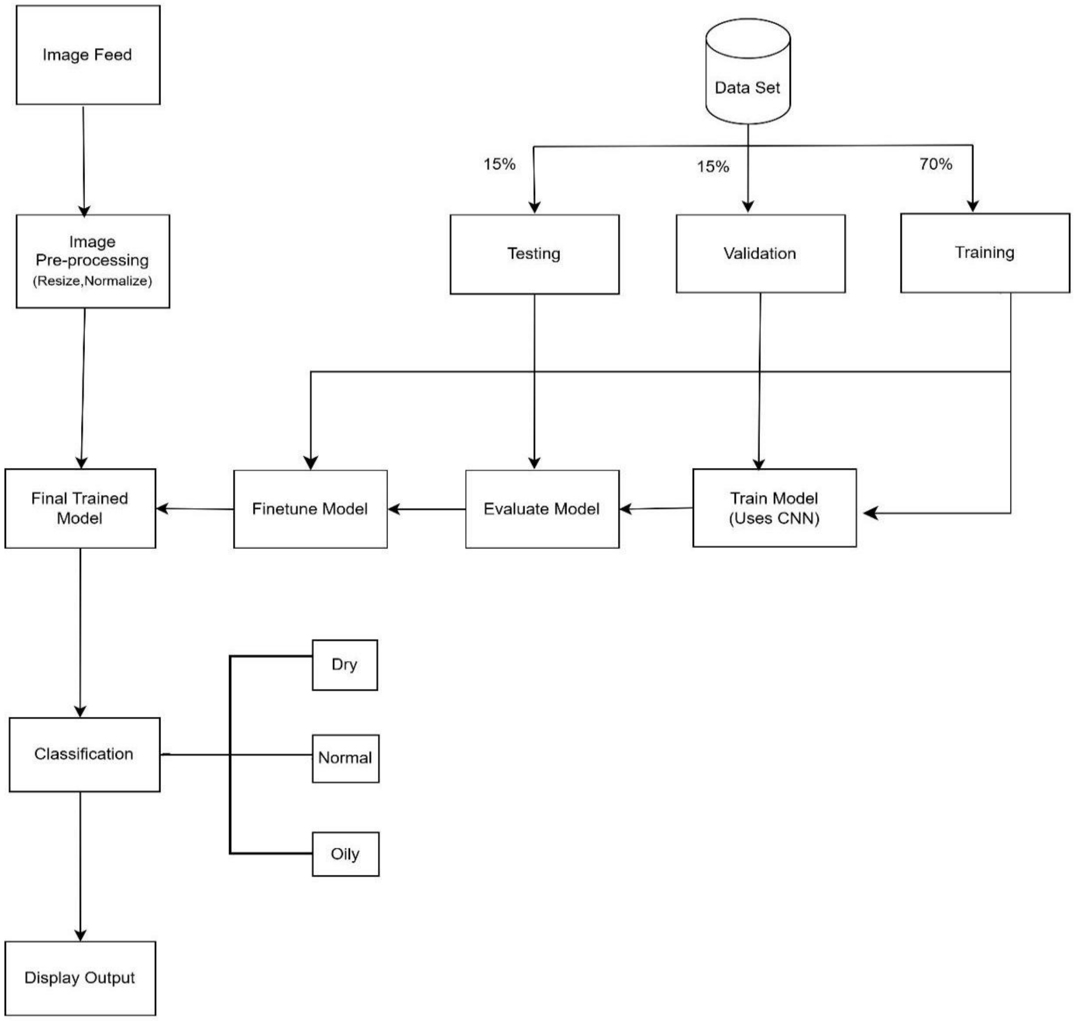


Figure 3.1. Model Training Diagram

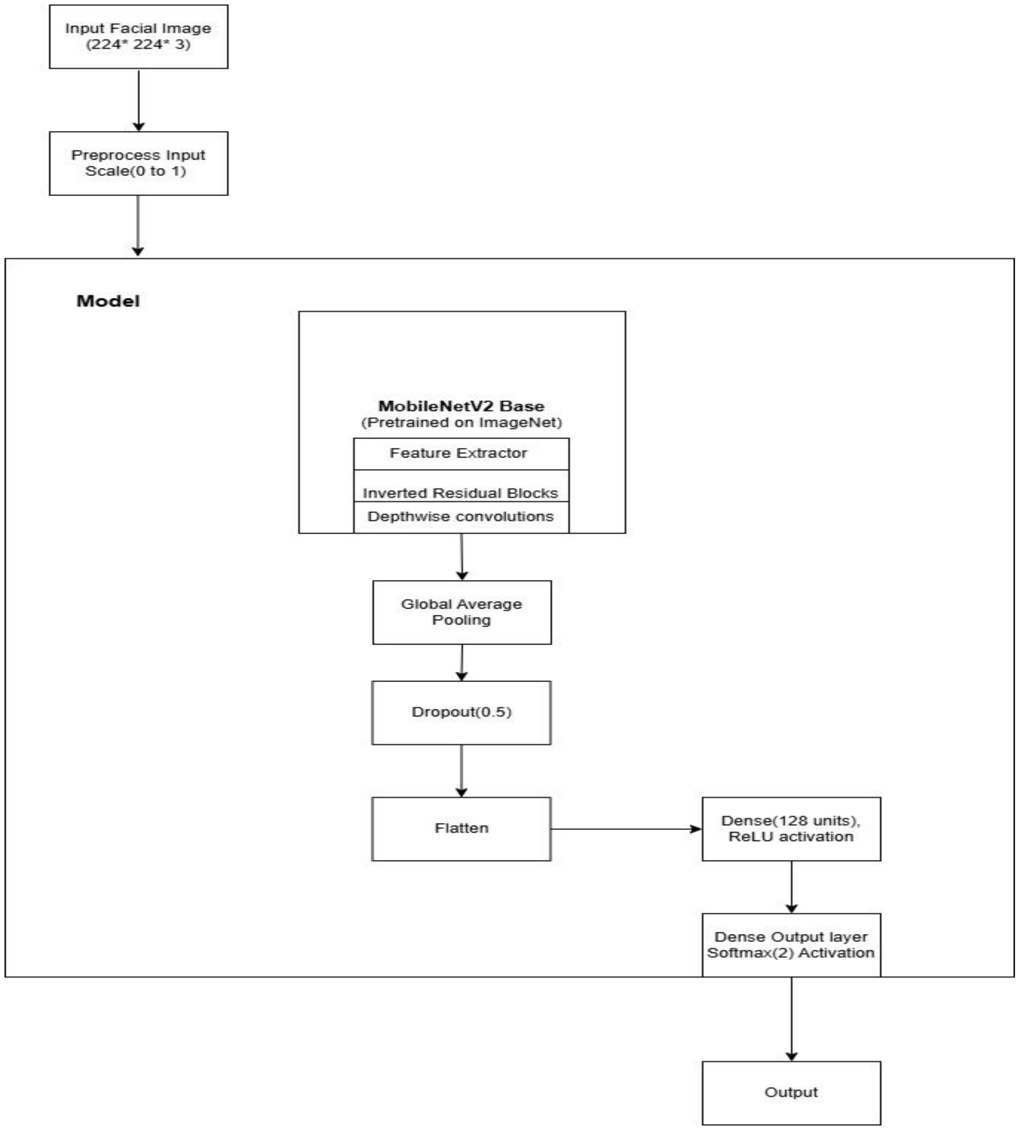


Figure 3.2: MobileNetv2 Training Process

### 3.1.1. Components of Model Training Diagram

**1. Data Set Collection**

The dataset for skin type detection will be gathered from various sources, including publicly available datasets and from Kaggle, which offers datasets specifically labeled for oily, dry, and normal skin types. The dataset will act as the "textbook" for the model, providing examples from which it will learn to recognize skin types. Large, diverse, and well-labeled datasets will enable the model to generalize well beyond the training data, reducing errors and improving performance on real world inputs.

**2. Data Pre-processing**

i. **Data Cleaning**

The collected facial skin images were cleaned by removing blurred, duplicate, poorly lit, and incorrectly labeled images. Through this process, better quality and relevant images were retained for training the model, which helped in learning accurate skin texture and features.

ii. **Image Resizing:**

All the cropped facial images were resized to a fixed dimension of 224×224 pixels to satisfy the input requirements of the MobileNetV2 model. This ensured uniformity across the dataset and enabled efficient processing during model training.

iii. **Normalization:**

To improve training efficiency, the image pixel values were normalized by scaling them from the range of 0–255 to 0–1. This helped stabilize the learning process and improve the convergence speed of the neural network.

iv. **Data Augmentation:**

Since the available dataset was limited, data augmentation techniques such as rotation, horizontal flipping, and zooming were applied. These transformations increased data diversity, reduced overfitting, and improved the model’s ability to generalize unseen facial images.

v. **Data Splitting:**

Finally, the preprocessed dataset was split into training, validation, and testing sets in an appropriate ratio. This allowed effective model training, performance tuning during validation, and unbiased evaluation of unseen data.

## 3.2. Overview of the System

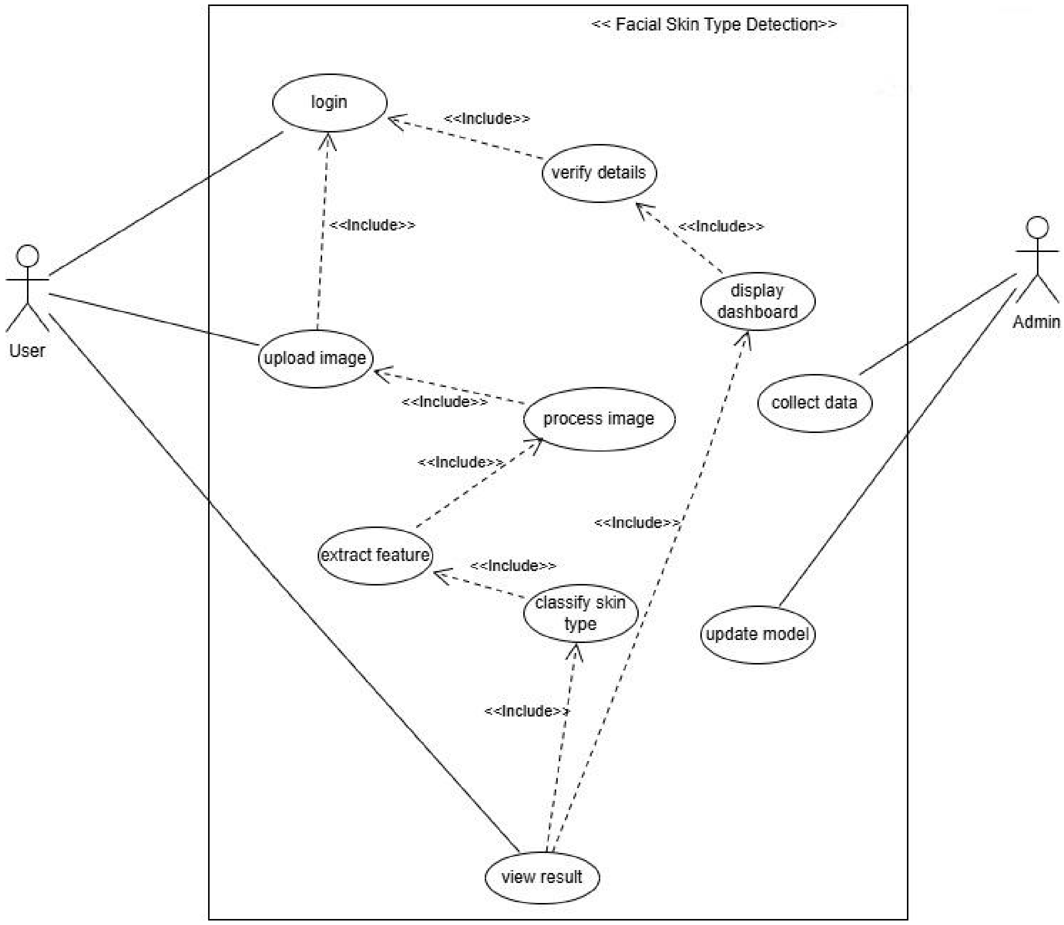


Figure 3.3: Use-Case Diagram

## 3.3. Software Development Life Cycle

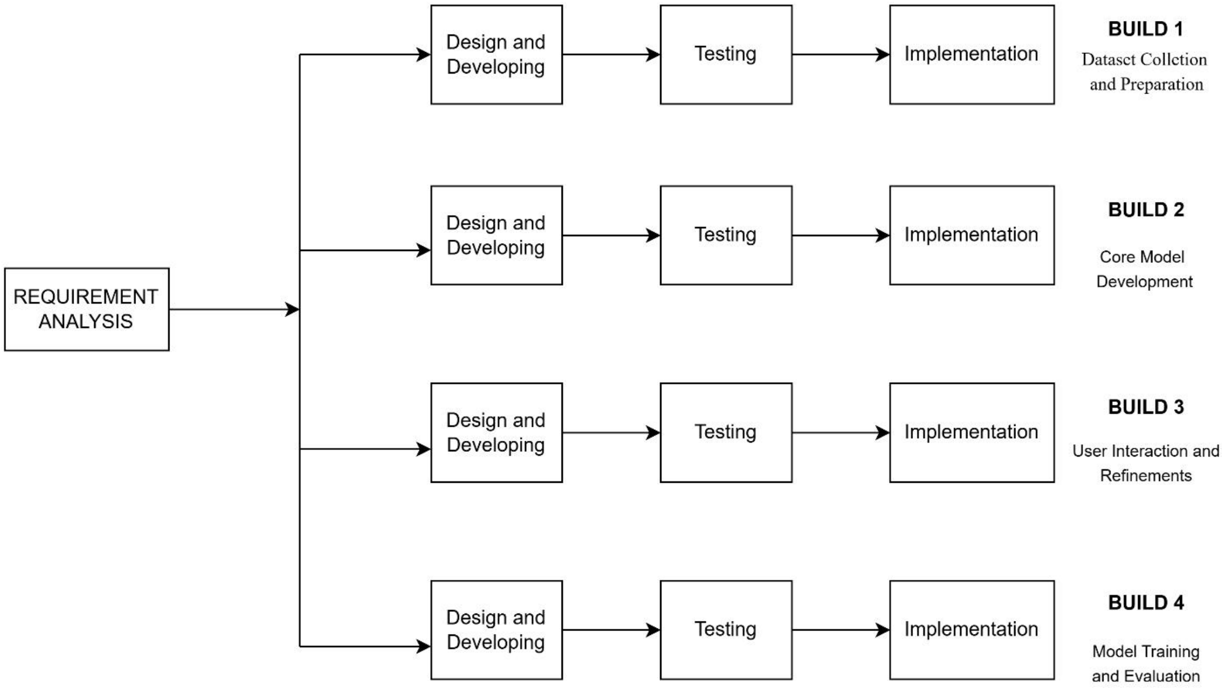


Figure 3.4: Software Development Lifecycle

# CHAPTER 4

# THEORETICAL FOUNDATION

## 4.1 Model Architecture

The proposed model architecture is based on a CNN using transfer learning with the MobileNetV2 architecture. The model is designed to automatically extract discriminative facial skin features from images and classify them into three categories: dry, normal, and oily. So basically, the architecture consists of two main components, namely as a feature extractor and a classifier.

i. **Input Layer**: Each facial image is preprocessed by resizing it to 224 × 224 pixels with three RGB channels and normalizing the pixel values. This ensures consistent input size and scale, enabling MobileNetV2 to process facial features efficiently and accurately.

ii. **Feature Extraction Module**: The model uses MobileNetV2 as a feature extractor, a lightweight CNN pre-trained on the ImageNet dataset. The convolutional layers are frozen to retain learned knowledge and reduce overfitting on the small dataset. MobileNetV2's depth-wise separable convolutions ensure low computational cost while maintaining strong feature extraction, making it well suited for facial skin type analysis.

iii. **MobileNetV2 Architecture:** MobileNetV2 is a lightweight and efficient CNN pre-trained on the ImageNet dataset. It uses depth-wise separable convolutions to reduce computational cost and model size while maintaining strong feature extraction performance.

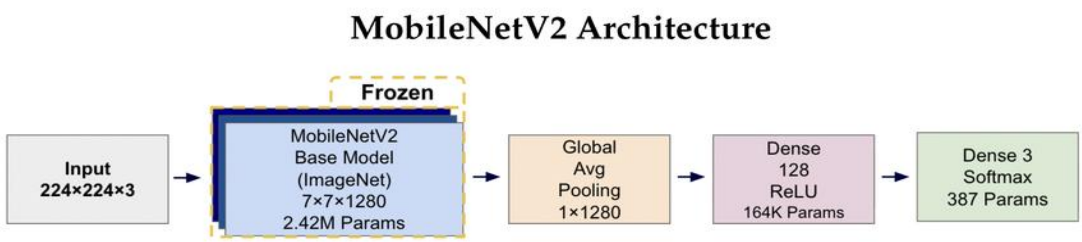


Fig 4.1: MobileNetV2 Architecture

Source: https://share.google/mvJUGo6WD5tQEDDel

iv. **Depth-wise Separable Convolution:** Depth-wise separable convolution in MobileNetV2 splits standard convolution into two steps: **depth-wise**

**convolution** extracts spatial features from each channel, and **pointwise (1×1) convolution** combines information across channels. This reduces computations and parameters, speeds up the model, and helps prevent overfitting, especially with small datasets.

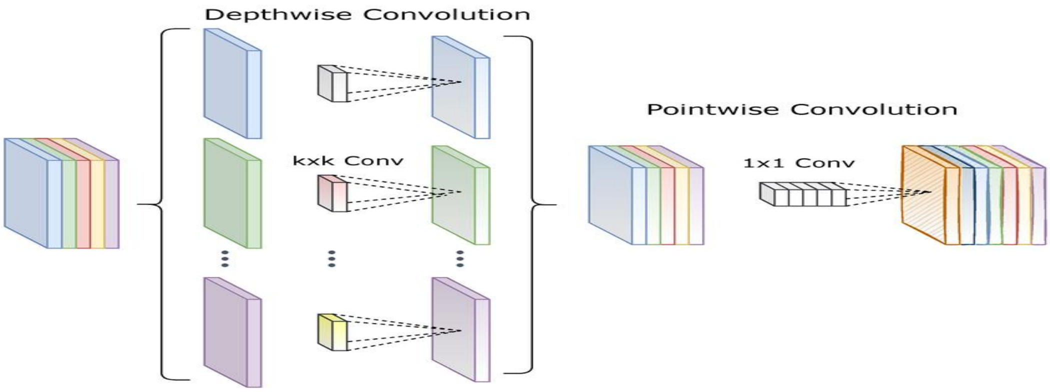


Fig 4.2: Depth wise Convolution and Pointwise Convolution.

Source: [https://share.google/O9tYc4oFK2VYo1IkQ](https://share.google/O9tYc4oFK2VYo1IkQ)

v. **Freezing the Base Model:** Freezing the base model keeps MobileNetV2's pre-trained weights unchanged, preserving learned features like edges and textures. This allows the model to focus on training only the new skin-type classification layers, reducing overfitting, and improving training efficiency on a small dataset.

vi. **Global Average Pooling layer:** After feature extraction, a Global Average Pooling (GAP) layer is used to reduce high-dimensional feature maps. GAP averages each feature map into a single value, producing a compact 1D feature vector that lowers model complexity, reduces overfitting, and improves classification efficiency.

vii. **Classification Module:** The classification module maps feature from MobileNetV2 to skin types (dry, normal, oily). The extracted feature maps are flattened into a 1D vector and passed through a Dense layer with ReLU to learn complex patterns. A Dropout layer reduces overfitting, and the final Dense layer with SoftMax outputs class probabilities, with the highest value giving the predicted skin type.

## 4.2 Model Evaluating

To evaluate the model, we feed the testing dataset which is 15% of the entire data which has already been split before. The model is then evaluated based on the evaluation of metrics, mainly test accuracy and testing loss. Furthermore, Classification Report, consisting of other evaluation metrics, are also generated.

Evaluation Matrices:

**i. Precision**

Precision measures how many of the predicted positive samples are correct.

$$\text{Precision}=\frac{TP}{TP+FP} \tag{4.1}$$

**ii. Recall (Sensitivity)**

Recall measures how many actual positive samples were correctly identified.

$$\text{Recall}=\frac{TP}{TP+FN} \tag{4.2}$$

**iii. F1-Score**

F1-score is the harmonic mean of Precision and Recall, providing a balanced measure.

$$\text{F1-score}=2\times\frac{Precision\times Recall}{Precision+Recall} \tag{4.3}$$

**iv. Accuracy**

Accuracy measures the overall correctness of the mode

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN} \tag{4.4}$$

**v. Confusion Matrix**

A confusion matrix summarizes prediction results by comparing actual and predicted classes

| **Actual/ Predicted** | **Positive** | **Negative** |
|---|---|---|
| Positive | TP | FN |
| Negative | FP | TN |

For

multi-class classification (dry, normal, oily), a confusion matrix is extended to a 3×3 matrix, where each row represents the actual class, and each column represents the predicted class

# CHAPTER 5
# RESULT AND DISCUSSION

## 5.1 Model Training and Evaluation

The training process for the skin type classification task utilizes a Transfer Learning strategy based on the MobileNetV2 architecture. Initially, the base model layers are frozen to preserve the pre-trained weights from the ImageNet dataset, allowing the newly added classification head to learn task-specific patterns such as skin texture and tone without disturbing general visual features.  The final model is evaluated using a hold-out test set to generate a Confusion Matrix and a Classification Report, ensuring the system generalizes well to unseen data.

**Validation Loss :** 0.5070 **Testing Loss :** 0.1976

**Validation Accuracy :** 0.8588 **Testing Accuracy :** 0.9309

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| Dry | 0.93 | 0.91 | 0.92 |
| Normal | 0.90 | 0.92 | 0.91 |
| Oily | 0.92 | 0.91 | 0.92 |
| Accuracy | | | 0.92 |
| Macro Avg | 0.92 | 0.92 | 0.92 |
| Weighted Avg | 0.92 | 0.92 | 0.92 |

Table 5.1: Classification Report

**Confusion Matrix**

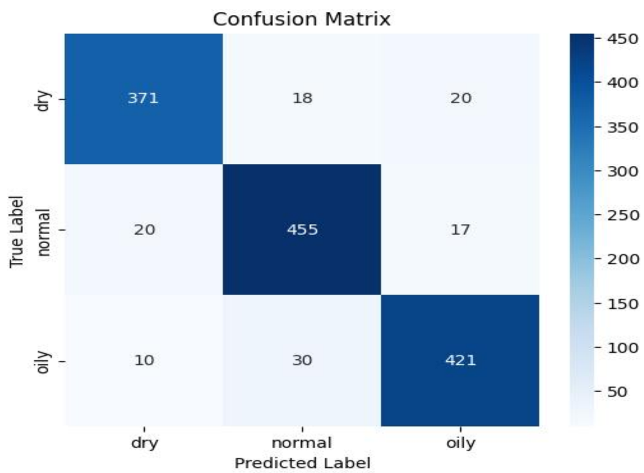


Figure 5.1: Confusion Matrix

**Observation while training the model**

Both the training and validation accuracy increased sharply during the first few epochs i.e form epochs 1 - 7. This indicated that the MobileNetV2 architecture is effectively leveraging its pre-trained ImageNet weights to quickly learn the specific features of "Oily," "Dry," and "Normal" skin types. The curves for training and validation are close to each other which might indicate good generalization. There are some fluctuations in the later training as it is normal during the training process. It suggested that the model is improving and adjusting.

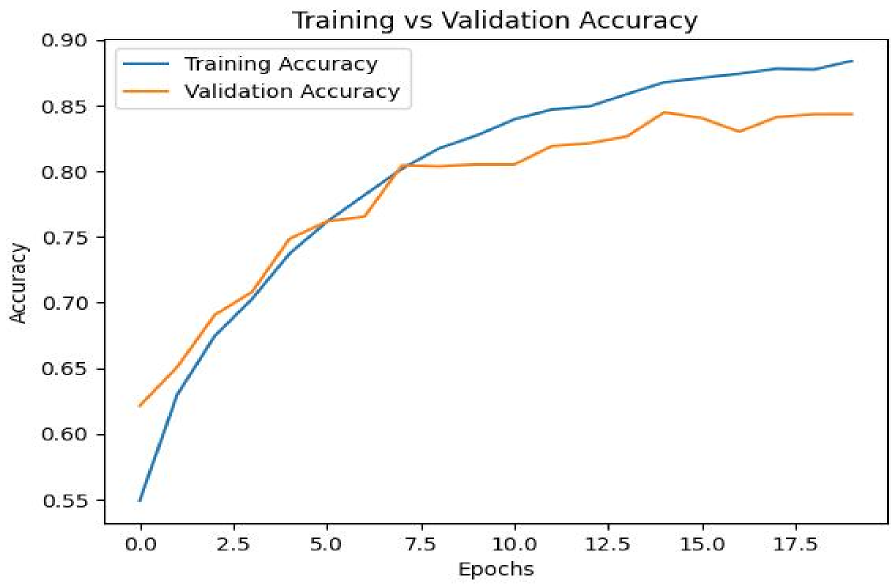


Figure 5.2: Training vs Validation Accuracy

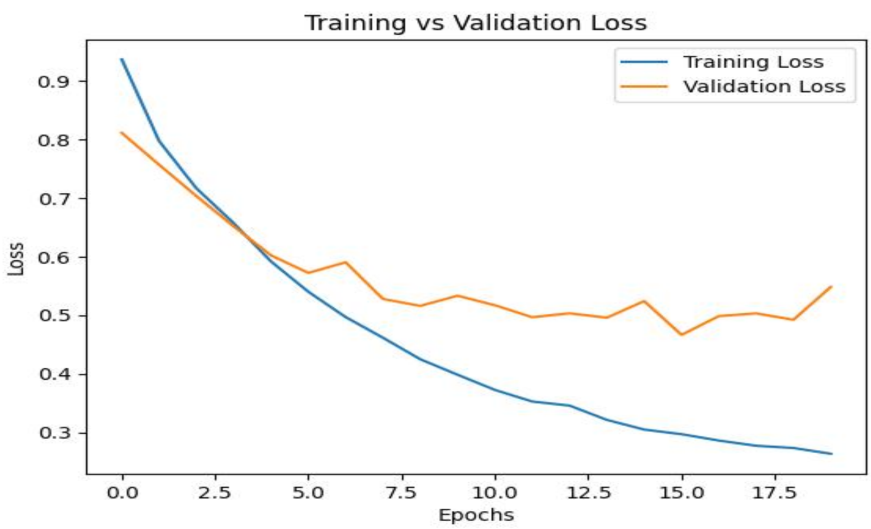


Figure 5.3: Training vs Validation Loss

### 4.1.7 Fine-Tuning the Model

Following the initial training phase, the MobileNetV2 model had under-gone fine-tuning to further specialize its feature extraction capabilities for skin type classification. This strategy involved 'unfreezing' a specific subset of the base model typically the last 20 to 30 convolutional layers allowing the network to adapt its higher-level layers to represent intricate, skin-specific features such as texture and oiliness while retaining the foundational knowledge gained from the ImageNet dataset. To preserve these pre-trained weights and ensure stable convergence, a significantly lower learning rate is applied, facilitating gradual weight adjustments without disrupting the model's prior learning. Throughout this phase, performance is rigorously monitored across both training and validation datasets to prevent overfitting and ensure the model generalizes effectively to unseen skin samples.

### 4.1.8 Training the Final Model.

After setting up the fine-tuning process, the model is then trained using the unfrozen layers of MobileNetV2 to help classify images better, and combining this fine-tuned network with the classification layers. The goal for this training phase was to give the network an enhanced ability to identify differences between very similar images.

## 5.2 Discussion

The Facial Skin Type Detection project was about making a system that uses intelligence to figure out what kind of facial skin people have, like dry skin, normal skin or oily skin just by looking at pictures. To do this, the team first had to get a lot of pictures of faces with skin colors, lighting and facial features. This was a deal because things like how bright or dark it is, the angle of the face and the texture of the skin can really affect how well the system works. And also made sure to get pictures in all sorts of conditions like in a studio and outside to make the system better at recognizing skin types in life.

When it came to building the system they used something called Convolutional Neural Networks or CNNs for short because they are really good at looking at pictures and finding patterns like how oily or dry someones skin's. It made the system simple and efficient so it could work well on websites or phones. It also made sure the pictures were all the size and quality so the system could understand them.

There were some problems in process of  building the system. Sometimes the lighting or the acne prones would trick the system for people with combination skin that is different in different areas.. Even with those problems the system was usually right when it came to people with really dry or really oily skin. It made the system in a way that would be easy to add features later like spotting acne or looking at skin color more closely.

What the learning  from this project is that the need a lot of pictures to make a system like this work and that CNNs are really good at looking at pictures and figuring things out. You also have to make sure the system is not too complicated or it will be slow. The project showed that you can make a system that's easy to use and helps people figure out their skin type without just guessing. The system is good. Can be used in the future to help people take better care of their skin as long as the pictures are good and the system is used in the right way. The Facial Skin Type Detection project is a start, for making systems that can help people with their skin.

# CHAPTER 6
# CONCLUSION AND FUTURE RECOMMENDATIONS

## 6.1 Conclusion

MUHAAR is a facial skin type detection system developed using deep learning techniques, using a transfer learning approach with the MobileNetV2 architecture to classify skin into dry, normal, and oily categories. The system is designed to provide a quick and accessible alternative to manual skin analysis.

During development, one of the minor challenges faced was the limited and less diverse dataset collected mainly from Kaggle. To address this, data augmentation techniques such as rotation, flipping, and zooming were applied to increase data diversity and improve the model's ability to generalize. Another challenge was the difficulty in distinguishing between similar skin types, especially normal and oily, due to overlapping features. To reduce this issue, proper preprocessing steps such as data cleaning, resizing, and normalization were implemented to ensure consistent input quality. Additionally, transfer learning with MobileNetV2 was used to take advantage of pre-trained features, and evaluation metrics like accuracy, precision, recall, and confusion matrix were used to monitor and improve performance.

In conclusion, MUHAAR is an effective and reliable system for skin type detection that reduces dependency on manual observation. Although some challenges were encountered during development, appropriate solutions such as data augmentation, preprocessing, and transfer learning were applied successfully, resulting in a system with good performance and potential for further improvement.

## 6.2 Future Recommendations

Future enhancements will focus on overcoming challenges like limited dataset diversity, difficulty distinguishing similar skin types, and variations in real-world images. The goal is to improve the system's accuracy, robustness, and usability, making it more reliable, scalable, and effective for practical applications.

1. Increase the dataset with diverse images from different skin tones, lighting conditions, and devices.
2. Fine-tune the model by unfreezing upper layers and training with a lower learning rate.
3. Extend the system to detect additional skin conditions like acne, pigmentation, and wrinkles.
4. Implement real-time detection using camera integration for interactive use.
5. Incorporate personalized s kincare recommendations based on detected skin type.
6. Deploy the system as a web or mobile application for wider accessibility.
7. Improve preprocessing techniques, such as illumination correction and region-of-interest extraction, to enhance accuracy.

# REFERENCES


[1] G. D. F. Y. L. L. Y. W. Jianghong Ran, "Automatic Measurement of Comprehensive Skin Types Based on Image Processing and Deep Learning," *Electronics 2025,* vol. 14, p. 21, 2024.

[2] S. P. Ruchika Chouhan, "A Fine Tune CNN Model for Human Skin Type Classification," *SSRG International Journal of Electrical and Electronics Engineering,* vol. 11, no. 11, p. 11, 2024.

[3] S. A. L. M. W. P. N. A. Sirawit Saiweo, "Human Skin Type Classification using Image Processing and Deep Learning Approaches," *Heliyon,* vol. 9, no. 11, p. 17, 2023.

[4] R. K. S. S. Ç. Fatma Bettil Kara Ardıç, "Skin Type Detection with Deep Learning: A Comparative Analysis," *Düzce University Journal of Science and Technology,* vol. 11, p. 14, 2023.

[5] H. Y. S. K. P. C. L. J. L. M. S. Y. Jinhee Lee, "Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions," *Journal of Cosmetic Dermatology,* vol. 23, no. 6, p. 12, 2021.

# ANNEX

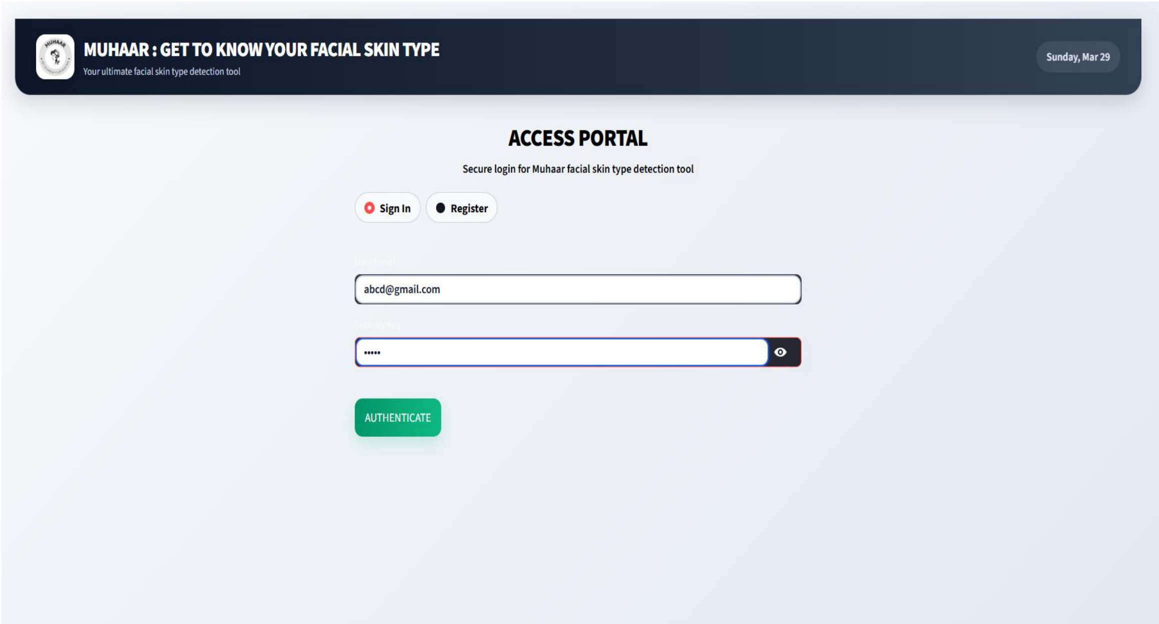


Figure 6.1 : Login Portal

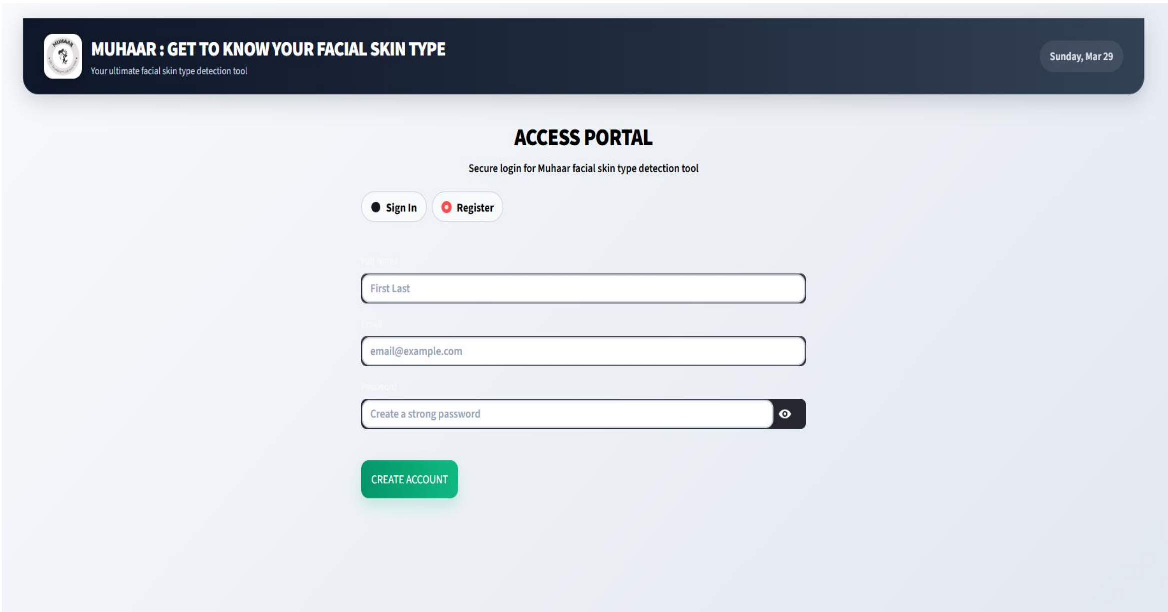


Figure 6.2 : Registration Portal

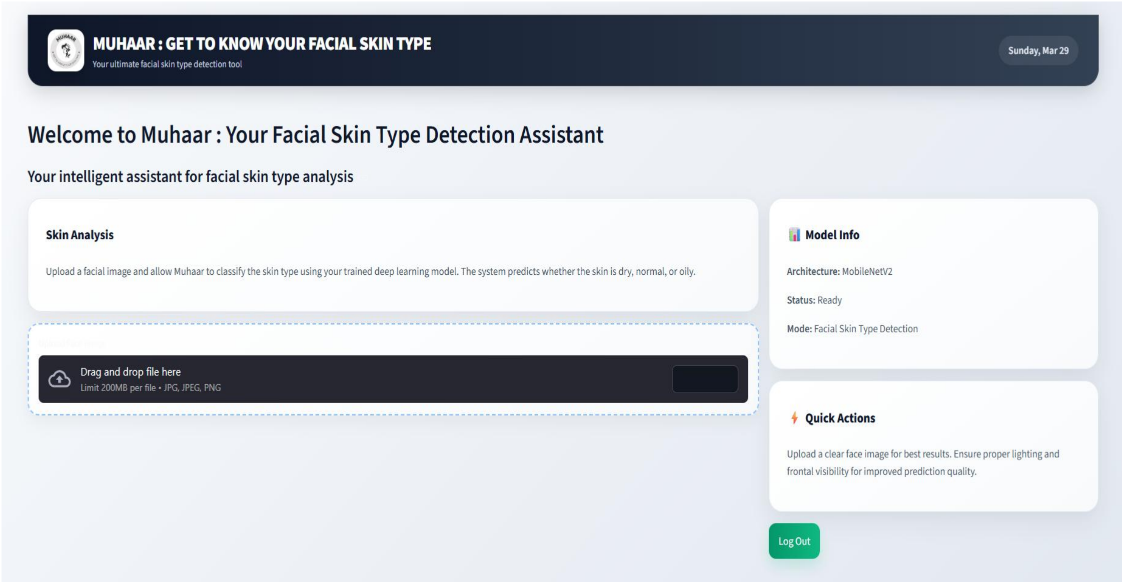


Figure 6.2 : Home Page

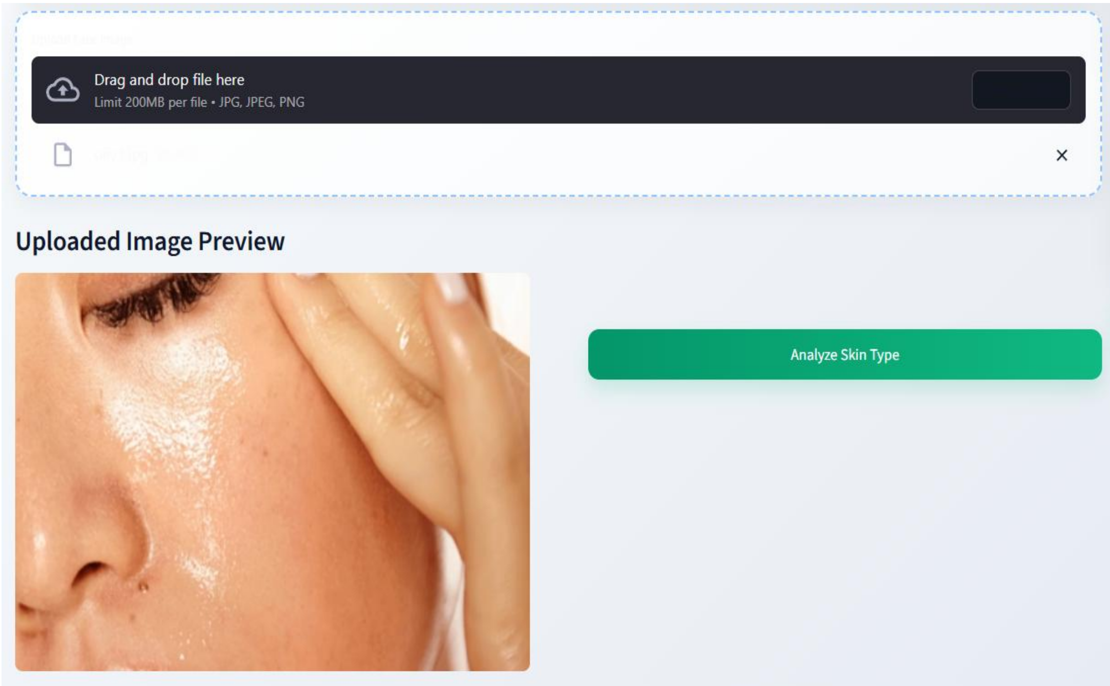


Figure 6.3 : Uploading a Facial Image

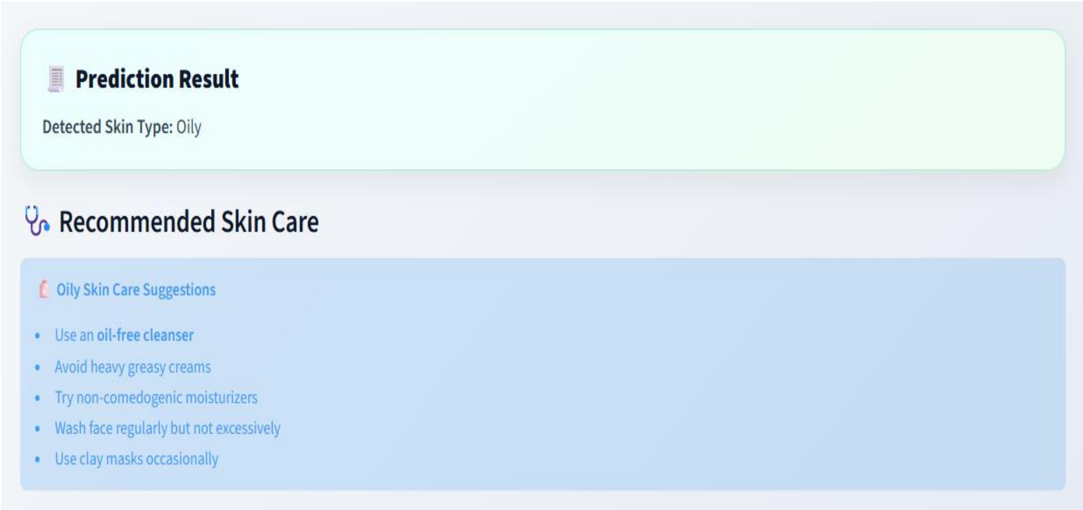


Figure 6.4 : Prediction Result